Deep Learning for Text Analytics

Welcome and Organization

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Course Format

- **Master level course with 6 ECTS**
  - Target group: Everybody with interest in advanced machine learning and textual data analysis
  - Mandatory elective or elective module depending on study program
- **Lecture (2 SWS) – Thursday 10:15 – 11:45 , room 202**
  - Introduction and discussion of relevant concepts
  - Lecture recordings will be shared via Moodle
- **Exercise (2 SWS) – Thursday 12:15 – 13:45, room 22**
  - Revisiting lecture concepts through Python demos
  - Working on programming exercises

Materials and Support for Self-Study

All material shared via Moodle (no enrollment key) / GitHub

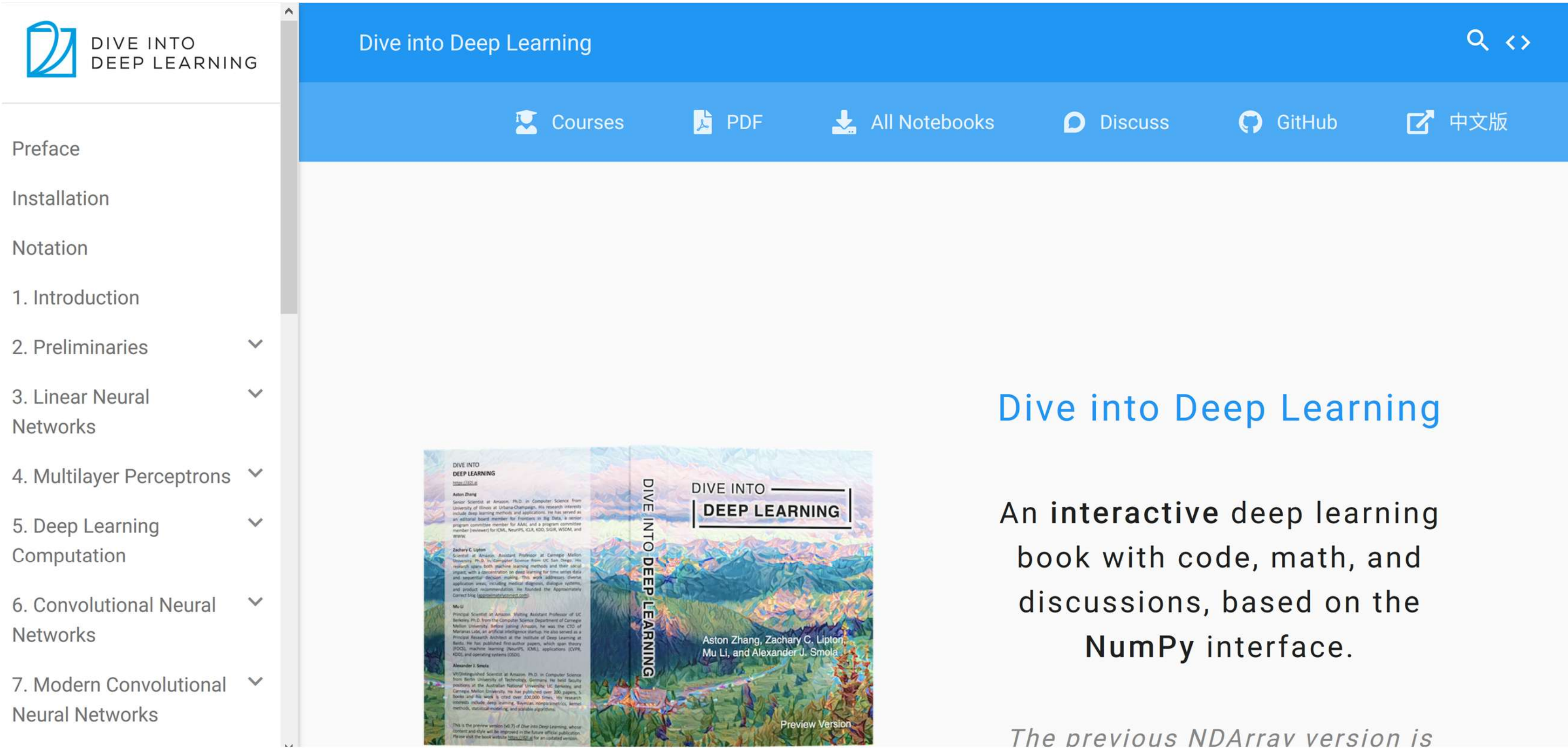
https://moodle.hu-berlin.de/course/view.php?id=140923

https://github.com/Humboldt-WI/delta

Further Materials and Support for Self-Study

The course draws on several books/papers/blogs/videos, etc.

- **Many great DL resources exist and are freely available online**
- **My favorite**
  - Covers everything you would want to know
  - Perfect mix of math and code demonstrations via notebooks



http://d2l.ai/index.html

Prerequisites

We assume you are familiar with standard ML workflows (i.e., BADS)

- **Assumed machine learning knowledge**
  - Foundations of supervised learning (regression vs. classification, loss functions)
  - Standard ML practices (e.g., cross-validation, hyperparameter tuning, ROC analysis)
  - Basic ML theory (bias, variance, overfitting)
- **Assumed practical skills**
  - Working knowledge in Python programming and associated tools (Jupyter, GitHub, etc.)
    - Use your own computer or cloud platform (HUB JupyterHub, Google Colab, AWS Sagemaker, etc.)
    - Our Python setup guide also provides some ideas (see Moodle)
  - Handling of tabular data sets using Numpy and Pandas
  - Foundations of data visualization for EDA
  - Coding ML workflows using sklearn

Portfolio Evaluation

Three tasks each awarding 0–100 points. Weighted average gives final score

		Scoring tableau			
		below expectations	meets expectations on most accounts	fulfills expectations on all accounts	significantly overfulfills expectations
		0 - 49	50-64	65-90	91-100
Task 1	**25%**				
Criteria a					
Criteria b					
Critera c					
Task 2	**35%**				
Criteria a					
Criteria b					
Critera c					
Task 3	**40%**				
Criteria a					
Criteria b					
Critera c					
Final score					

We calibrate the final grade on the score distribution

Portfolio Evaluation

Three tasks each awarding 0–100 points. Weighted average gives final score

- **You have to register for the portfolio**
  - Registration period 16.04.26 – 28.05.26
  - Cancel registration up to 28.05.26 if eventually deciding against taking the course
- **Format of the portfolio**
  - Task 1 (25%): Programming task
    - Demonstrate ability to develop a Keras-based DL model for tabular data, which we will provide
    - Deliverable: Jupyter notebook
    - Deadline: 29 May 2026
  - Task 2 (35%): Written exam (60 min)
    - Demonstrate your understanding of core DL / NLP concepts discussed in the lecture
    - Format: Mostly free-text questions with pre-defined answer space (e.g., explain in 3 sentences…)
    - Date: 16 July 2026 (last lecture session)
  - Task 3 (40%): Programming task concerning text classification
    - Deliverable : Jupyter Notebook
    - Deadline: 03 August 2026
    - This task has 3 sub-tasks designed such that you can start early and build your final notebook week-by-week

Course Outline & Schedule

Semester week				Lecture	Tutorial	Portfolio
1	13. Apr	-	17. Apr	Introduction	ML Basics (self-study)	Registration opens
2	20. Apr	-	24. Apr	Deep Learning (DL) Foundations	DL training & tuning	
3	27. Apr	-	01. Mai		The Keras library for DL	Ready for Task I
4	04. Mai	-	08. Mai		Q&A Programming Task I	Ready for Task II.a.
5	11. Mai	-	15. Mai	NLP Foundations	NLP: Text cleaning & dictionaries	
6	18. Mai	-	22. Mai	Word Embeddings	NLP: Word embeddings	
7	25. Mai	-	29. Mai	Recurrent networks (RNN)	RNN: Time Series forecasting	28.5. - Last chance to register 28.5. - Last chance to unregister 29.5. - Submission Task 1
8	01. Jun	-	05. Jun	Advanced RNNs	RNN: Text classification	
9	08. Jun	-	12. Jun	NLP Transfer learning	Q&A Programming Task II	Ready for task II.b.
10	15. Jun	-	19. Jun	Attention Mechanism	Attention Mechanism	
11	22. Jun	-	26. Jun	Transformer Architecture I	Transformers for text classification	Ready for task II.c
12	29. Jun	-	03. Jul	Transformer Architecture II	More on transformers & LLMs	
13	06. Jul	-	10. Jul	Summary & Outlook	Q&A Written Exam	
14	13. Jul	-	17. Jul			16.5. - Written exam
	03. Aug	-	07. Aug			03.8. - Submission Task 2

Credits & Disclaimer

- **The materials provided in the course have been inspired and draw upon numerous great sources available online**
- **The authors acknowledge the use of and wish to express their gratitude to the following sources**
  - UC Berkeley STAT 157: Dive into Deep Learning (http://d2l.ai/index.html)
  - MIT 6.S191: Introduction to Deep Learning (http://introtodeeplearning.com/)
  - Stanford CS224n: Natural Language Processing with Deep Learning (https://web.stanford.edu/class/cs224n/)
  - Oxford University: Deep Natural Language Processing course (https://github.com/oxford-cs-deepnlp-2017)
  - N. d. Freitas Deep Learning at Oxford 2015 (https://www.youtube.com/playlist?list=PLE6Wd9FR--EfW8dtjAuPoTuPcqmOV53Fu)

Thank you for your attention!

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